

RESEARCH NOTE

Treating unobserved heterogeneity in PLS path modelling: A comparison of FIMIX-PLS with different data analysis strategies

Marko Sarstedt^{a*} and Christian M. Ringle^b

^a*Institute for Market-Based Management, University of Munich, Munich, Germany;* ^b*Institute for Industrial Management, University of Hamburg, Hamburg, Germany*

(Received 22 April 2008; final version received 6 May 2009)

In the social science disciplines, the assumption that the data stem from a single homogeneous population is often unrealistic in respect of empirical research. When applying a causal modelling approach, such as partial least squares path modelling, segmentation is a key issue in coping with the problem of heterogeneity in the estimated cause–effect relationships. This article uses the novel finite-mixture partial least squares (FIMIX-PLS) method to uncover unobserved heterogeneity in a complex path modelling example in the field of marketing. An evaluation of the results includes a comparison with the outcomes of several data analysis strategies based on *a priori* information or *k*-means cluster analysis. The results of this article underpin the effectiveness and the advantageous capabilities of FIMIX-PLS in general PLS path model set-ups by means of empirical data and formative as well as reflective measurement models. Consequently, this research substantiates the general applicability of FIMIX-PLS to path modelling as a standard means of evaluating PLS results by addressing the problem of unobserved heterogeneity.

Keywords: partial least square (PLS); path modelling; heterogeneity; latent class; finite mixture; market segmentation; corporate reputation

1. Introduction

Applications of structural equation models (SEMs) are usually based on the assumption that the analysed data stem from a single population. In many real-world applications, this assumption of homogeneity is often unrealistic, as individuals are likely to be heterogeneous in their perceptions

*Corresponding author. Email: sarstedt@bwl.lmu.de

and evaluations of latent constructs. Since its formal introduction in the 1950s, market segmentation has been one of the primary marketing ideas in respect of product development, marketing strategy, and understanding customers. However, the true distribution of heterogeneity can never be known *a priori*, and thus, there are cases where it is difficult to find homogeneous customer segments. The development of analytic methods with which to segment markets has lagged behind their need in business applications [36]. Whereas modelling heterogeneity in covariance structure analysis (CSA; [24]) has been studied for several years, research interest has only lately turned to the question of clustering in partial least squares path modelling (PLS-PM) [21]. Recently, proposed PLS-PM multi-group analysis methods with which to compare data derived from *a priori* information and PLS-PM segmentation [6,9,20], as well as newly developed PLS-PM clustering techniques [42], such as a latent class approach, generally hold considerable promise for market segmentation studies.

Latent class modelling [25] is a useful clustering tool for uncovering groups of individuals with similar preferences and sensitivities. In CSA, Jedidi *et al.* [24] pioneered this field of research and proposed the finite-mixture SEM approach that treats heterogeneity and forms data segments in the context of a specified model structure in which all manifest variables are measured with error. Although the original technique extends CSA, and is implemented in software packages for statistical computations, such as Mplus [29], it is inappropriate for PLS-PM's dissimilar methodological assumptions [14]. Consequently, Hahn *et al.* [19] introduced the finite-mixture partial least squares (FIMIX-PLS) method that specifically copes with the ordinary least squares-based PLS-PM predictions when combining a finite-mixture procedure with an expectation–maximization (EM) algorithm.

Building on the guiding articles by Jedidi *et al.* [24] and Hahn *et al.* [19], this article presents the FIMIX-PLS method as recently implemented in the statistical software application SmartPLS [38] and, thereby, made broadly applicable to social science research as a primary approach with which to segment data based on PLS-PM results. The FIMIX-PLS approach allows model parameters to be estimated and observations' affiliations to be simultaneously segmented. Even though FIMIX-PLS has been the key PLS-PM segmentation approach to date, it has seldom been used on marketing examples with typical heterogeneous data. Ringle [33] has presented one of the few empirical studies in this regard. However, the author applies a very sparse model design, which raises the question of how the method will perform in more complex model set-ups with a greater number of constructs and indicators. Furthermore, the empirical application does not account for formative measurement models, which are becoming increasingly relevant in respect of SEM applications, such as success factor studies in marketing, and for the estimation of which PLS-PM has proven to be particularly advantageous [1].

The primary contribution of this article is, initially, to evaluate the effectiveness of FIMIX-PLS-PM in a complex application and to compare its performance when using *a priori* information and available segmentation strategies, such as *k*-means, for both manifest and latent variable scores. Methodological considerations that are relevant to a comparison of above described models include an assessment of item and construct reliability as well as determination model adequacy. Since the primary concern of the FIMIX-PLS algorithm is to capture heterogeneity in the inner model, the focus of the comparison is the evaluation of the inner model's goodness of fit (GoF) as reflected in the “variance accounted for” by the endogenous latent variables. Nevertheless, several measures are computed to assess the constructs' reliability and validity, which is a prerequisite for deriving meaningful solutions from any structural equation modelling analysis. The differences between the path coefficient estimates are analysed for significance by means of Henseler's [20] newly developed PLS-PM multi-group comparison procedure.

The applicability of FIMIX-PLS is a critical issue to facilitate its usage and stimulate ongoing development in this important research field. By applying FIMIX-PLS to empirical data to test the method's capability to identify and treat unobserved heterogeneity in PLS path models, this

101 article answers the call of previous studies by Esposito Vinzi *et al.* [12] as well as Ringle [33]. This contribution is of importance for both researchers and practitioners, as it also points out the need to capture heterogeneity in SEM analyses.

The remainder of the article is organized as follows: the next section deals with the available approaches for handling heterogeneity in PLS-PM and introduces the FIMIX-PLS methodology in more detail. Thereafter, a marketing-related empirical example is presented, which serves as the basis for the utilization of various different data analysis strategies. Subsequently, the results of the analyses are presented with regard to both the evaluation of measurement models and that of the inner model. The article concludes with a discussion of the implications of the findings in respect of PLS-PM and by offering directions for further research.

2. Clustering in PLS-PM

116 Researchers frequently start sequential procedures in which homogenous subgroups, based on *a priori* information, are formed to account for heterogeneity, or they revert to applying cluster analysis techniques to data, followed by multi-group structural equation modelling [6,9,20]. However, none of these approaches is generally considered satisfactory, as observable characteristics, such as gender, age, or usage frequency, are often insufficient to capture heterogeneity adequately [56]. Conversely, the application of traditional cluster analysis techniques suffers from conceptual shortcomings and cannot account for heterogeneity in the relationships between latent variables [24]. Different approaches, designed to capture and treat unobserved heterogeneity in path models, have therefore been proposed to detect latent classes. These procedures generalize, for example, tree-like structure [40], finite mixture [19], fuzzy regression [31], distance-based [13,49] and genetic algorithm approaches [34,37] to PLS-PM. So far, only FIMIX-PLS has been comprehensively evaluated in simulation studies [12,33,35] and made available to researchers through the SmartPLS software application [38]. Moreover, Sarstedt's [42] review of PLS-PM segmentation techniques characterizes FIMIX-PLS as the primary choice for segmentation tasks in PLS-PM.

The FIMIX-PLS approach combines the strengths of the PLS-PM method with the advantages of finite-mixture modelling when carrying out segmentation tasks in structural equation modelling. 131 A finite-mixture approach to model-based clustering assumes that the data originate from several subpopulations or segments [28]. Each segment is modelled separately and the overall population is a mixture of segment-specific density functions. Homogeneity is no longer defined in terms of a common set of scores, but at a distributional level. Thus, finite-mixture modelling enables marketers to cope with heterogeneity in data by clustering observations and estimating parameters simultaneously, thus avoiding well-known biases that occur when models are estimated separately [30]. While Hahn *et al.* [19] developed the FIMIX-PLS methodology, recent contributions include a systematic approach with which to apply the methodology [36], a software application [38], studies on artificially generated data [12,33,35], and some empirical applications [39,45,46].

The FIMIX-PLS approach keeps all measurement model parameters fixed along the analysis and uses a PLS path model estimation's latent variable scores on the aggregate data level to simultaneously compute the segment-specific model parameters in the inner model. Thereby, FIMIX-PLS ascertains the heterogeneity of the data structure within a PLS-PM framework. In FIMIX-PLS's first step, a path model is estimated by using the PLS-PM algorithm and empirical data based on manifest variables' values in the outer measurement models [19]. The resulting latent variable scores are then employed to run the FIMIX-PLS algorithm in the second step. 146 Hence, FIMIX-PLS only aims at capturing unobserved heterogeneity in the estimated inner model parameters in respect of the relationships between latent variables.¹

FIMIX-PLS computes the probabilities of membership for each observation to fit into the predetermined S numbers of segments. Each endogenous latent variable η_i is defined by the

151 weighted average of segment-specific distributional functions $f_{i|s}(\cdot)$,

$$\eta_i \sim \sum_{s=1}^S \pi_s f_{i|s}(\eta_i | \xi_i, \mathbf{B}_s, \mathbf{\Gamma}_s, \mathbf{\Psi}_s), \quad (1)$$

156 where ξ_i is an exogenous variable vector in the inner model in respect of observation i , $\mathbf{B}_s$ is the path coefficient matrix of the endogenous variables, and $\mathbf{\Gamma}_s$ of the exogenous latent ones, whereas $\mathbf{\Psi}_s$ depicts the matrix of each segment's regression variances of the inner model on the diagonal, zero else. The mixing proportion π_s determines the relative size of segment s ($s = 1, \dots, S$) with $\pi_s > 0 \forall s$ and $\sum_{s=1}^S \pi_s = 1$. Assuming that each vector of endogenous latent variable η_i is distributed as a finite mixture of conditional multivariate normal densities allows
161 $f_{i|s}(\eta_i | \xi_i, \mathbf{B}_s, \mathbf{\Gamma}_s, \mathbf{\Psi}_s)$ to be substituted and, thereby, changes Equation (1) into the following segment-specific distributional function:

$$166 \quad \eta_i = \sum_{s=1}^S \pi_s \left[\frac{|\mathbf{B}_s|}{(2\pi)^{J/2} \sqrt{|\mathbf{\Psi}_s|}} \right] \exp \left\{ -\frac{1}{2} (\mathbf{B}_s \eta_i + \mathbf{\Gamma}_s \xi_i)' \mathbf{\Psi}_s^{-1} (\mathbf{B}_s \eta_i + \mathbf{\Gamma}_s \xi_i) \right\}, \quad (2)$$

where J depicts the number of endogenous latent variables in the inner model. An EM formulation of the FIMIX-PLS algorithm is used for statistical computations to maximize likelihood and to ensure convergence in this model. By differentiating between dependent (i.e. endogenous latent) and explanatory (i.e. exogenous latent) variables in the inner model, the approach
171 follows a mixture regression concept [16]. This concept allows the estimation of separate linear regression functions and several segments' corresponding object memberships. Consequently, FIMIX-PLS provides the probabilities of each observation belonging to a specific segment s .

176 One difficulty with the application of FIMIX-PLS is related to the decision regarding how many segments to retain from the data. This decision is crucial for real-world applications, as many managerial decisions rely on it [43]. Various tests and heuristics have been proposed to determine the number of segments, but this so-called model selection problem is a longstanding and still unresolved issue with the least satisfactory statistical treatment.

181 An obvious way to determine the number of segments in the mixture model is to test the null hypothesis of S segments against the alternative hypothesis of $S + 1$ segments by carrying out a standard likelihood ratio test (LRT). However, as pointed out by several authors [28,56], the LRT is inappropriate in a finite-mixture framework, because the test statistic does not follow a central χ^2 -distribution under the null hypothesis. In light of this problem, researchers frequently revert to
186 a heuristic approach in the form of model selection criteria to determine the number of segments that can be classified as information and classification criteria.

Information criteria are based on a penalized form of the likelihood, as they simultaneously take a model's GoF (likelihood), and the number of parameters used to achieve that fit, into account. These criteria therefore correspond to a penalized likelihood function; that is, the negative likelihood plus a penalty term that increases with the number of parameters and/or the
191 number of observations. Information criteria generally favour models with large log-likelihoods and few parameters and are scaled so that a lower value represents a better fit. In applications, several competing models with alternating numbers of segments are examined and the model that minimizes the value of the information criterion is selected.

196 In keeping with substantive theory, applied researchers usually use a combination of criteria to guide their decision. Popular criteria include the Akaike information criterion (AIC), the modified AIC with factor 3 (AIC₃), the consistent AIC (CAIC), and the Bayes information criterion (BIC). Sarstedt *et al.* [47] have provided a comprehensive evaluation of these criteria in FIMIX-PLS-PM analyses. The results show that the CAIC is the best information criterion to use with a

large variety of data configurations, whereas the AIC (BIC) shows a pronounced tendency to overestimate (underestimate) the correct number of segments. Consequently, researchers should primarily revert to CAIC when deciding on the number of segments.

In addition to these criteria, the normed entropy statistic (EN), as proposed by Ramaswamy *et al.* [32], is a critical criterion for analysing whether segment-specific FIMIX-PLS results produce well-separated clusters. For example, as the targeting of markets requires the segments to be differentiable, they must be conceptually distinguishable to explain their different responses to alternative marketing-mix elements and programmes. EN criterion values may range between 0 and 1 and indicate the fuzziness of the partition based on the case-by-case estimated FIMIX-PLS probabilities of membership. An increase in the distinctiveness of the derived classes' separation is commensurate with an increase in the EN criterion. Applications of FIMIX-PLS provide evidence that EN values of 0.50 indicate well-separated class memberships that allow meaningful *a priori* segmentation of specific PLS-PM estimations. This kind of analysis allows establishing comprehensible interpretations of the results and forming sound managerial implications.²

In respect of avoiding unreasonable FIMIX-PLS results, a useful indicator is the small size of additional segments. Compared with alternative procedures (e.g. the Newton–Raphson method), the EM algorithm always converges to a pre-specified number of segments S . This characteristic of the EM algorithm is advantageous, but when the number of segments is over-specified, some observations are “forced” to belong to the extraneous segment even though they actually fit in another segment. In such situations, the additional segment is only small, which explains the marginal amount of heterogeneity in the overall data set.

A primary concern regarding FIMIX-PLS relates to the EM algorithm, which converges to local maxima. This problem generally seems to deteriorate when the number of estimated parameters is large, or the segments are not well separated [56]. The latter observation also applies to FIMIX-PLS. However, this risk is usually resolved by choosing different starting random values for the algorithm. Secondly, according to a strictly theoretical viewpoint, the imposition of a distributional assumption regarding the endogenous latent variable may prove to be problematic. However, recent simulation evidence shows the algorithm to be robust against violations of the distributional assumptions [12].

A drawback of the FIMIX-PLS methodology is related to the algorithm using the aggregate data level's latent variable scores as input for segmentation and, thereby, keeping the measurement model constant across all segments to ensure convergence. Consequently, FIMIX-PLS applies mixtures to the regressions in the inner model and only captures heterogeneity in these relationships. The newly developed response-based segmentation methods for PLS-PM, PLS-GAS [34,37], and REBUS-PLS [13] overcome this shortcoming by dynamically forming new data groups and computing group-specific outer and inner PLS path model estimates in every iteration to uncover heterogeneity in both the structural and the measurement models. Nevertheless, in FIMIX-PLS, each segment's regression equations determine the inner model relationships while the method simultaneously uncovers segments, thus reliably accounting for heterogeneity in latent variables' relationships. Ringle *et al.* [39] demonstrate this capability by means of two numerical experiments. These authors also show that FIMIX-PLS results ought not to be analysed and interpreted immediately.

In a subsequent analytical step [36], the ex-post analysis, an explanatory variable, which allows data groups to be formed as indicated by FIMIX-PLS, must be identified. These *a priori* segmented data are then used as new inputs for PLS-PM estimations, providing group-specific latent variables scores as well as results regarding the outer and inner measurement models. However, the FIMIX-PLS results are based on each observation's probabilities of class membership. Using an explanatory variable for the group-specific PLS-PM estimations in the ex-post analysis can never exactly match the FIMIX-PLS results. The goal is to explain the uncovered segments by matching

the ex-post results as well as possible with the FIMIX-PLS outcomes. However, the success of this important analysis step depends on the existence and quality of potential explanatory variables in the available data.

Moreover, the ex-post analysis is of the utmost importance, since measurement models can only be updated through this ex-post analysis, thus leading to segment-specific measurement model parameters. Accordingly, the criticism regarding the utilization of latent variable scores as input for the FIMIX-PLS algorithm is somewhat relaxed and turned into a key advantage of this segmentation approach: FIMIX-PLS is generally applicable to all kinds of PLS path models, regardless of whether the latent variables build on formative or reflective measurement models [39].

Formative measurement has become increasingly important for structural equation modelling [18] and is a key advantage of variance-based structural equation modelling [8]. Consequently, the applicability of FIMIX-PLS is a crucial aspect in this regard. The identification of heterogeneity in the inner model can be related to formative measurement models, which, in one extreme, can have homogenous weights structures – all outer weights are at comparable levels and there are no significant differences in the PLS-PM multi-group analysis – and, in the other extreme, can have highly heterogeneous outer weight relationships with significant group-specific differences. Regardless of the type of formative measurement that leads to heterogeneity in the inner model relationships, FIMIX-PLS uncovers this heterogeneity by means of latent variable scores and, thereby, identifies (e.g. in the ex-post analysis) data groups that may have homogenous or heterogeneous outer weight relationships [39]. The empirical study (Section 3) substantiates this finding by providing an example of these different kinds of occurrences, which further stresses the general applicability of FIMIX-PLS to PLS-PM.

In addition to PLS-GAS [34,37] and REBUS-PLS [13], FIMIX-PLS can currently be regarded as a key approach with which to capture heterogeneity in PLS-PM, despite some limitations [42]. The FIMIX-PLS algorithm has been integrated into the easy-to-use statistical software application SmartPLS, which can be used in various fields of marketing, as some initial studies [45,46] have illustrated. Ringle *et al.* [35] as well as Esposito Vinzi *et al.* [12] demonstrate FIMIX-PLS's ability to identify heterogeneity in the inner model by applying the methodology to a numerical example, using experimental data. Furthermore, Ringle *et al.* [39] demonstrate FIMIX-PLS's capability to reveal heterogeneity in path models with formative measures by conducting an ex-post analysis. However, the complexity of the model used in these studies is rather limited. Likewise, Ringle [33] offers a scarce application with empirical data by using a brand preference model to evaluate the effects of the two reflective constructs "price" and "quality" (measured by means of four indicators each) on the construct "satisfaction", which is operationalized by means of two reflective indicators. The very simple model is then used to compare the algorithm's performance with sequential clustering procedures. The results demonstrate that FIMIX-PLS is a useful methodology that successfully supplements traditional segmentation techniques. However, the question arises: how will the methodology perform in more complex model set-ups?

3. Application to corporate reputation data

3.1 Path model with latent variables and empirical data

The empirical example in this study extends Ringle's [33] study by applying FIMIX-PLS to a complex path model, which captures the effects of corporate-level marketing activities on the mediating construct "corporate reputation" and, finally, on "satisfaction" as well as "loyalty" in the German telecommunications sector [10]. Compared with previous studies, the increased number of constructs and manifest variables as well as the integration of both formative and reflective indicators mirrors actual applications in the consumer research community more accurately. This

exemplary model is also used to compare the performance of FIMIX-PLS with the utilization of *a priori* information to form segments and the *k*-means clustering methodology.

In accordance with an analysis by Eberl [10], the relationships between marketing activities and customer loyalty are strongly mediated by corporate reputation, which draws on a two-dimensional conceptualization [48]. One dimension comprises all cognitive evaluations of the company (“competence”), whereas the second dimension captures affective judgements (“likeability”). “Competence” and “likeability” draw on three indicators, which were identified as exchangeable and, hence, form a reflective measurement model. Past research identified four exogenous driver constructs of corporate reputation (“quality”, “performance”, “attractiveness”, and “corporate social responsibility” (CSR)), which have been shown to be robust across different data sets, countries, and industries [11,48]. All exogenous latent variables have a formative measurement and, hence, a total of 21 formative indicators measure the levers of corporate-level marketing activities. A complete list of all manifest variables is provided in Table 1. The two primary dimensions of corporate reputation relate to customer satisfaction, while likeability also influences customer loyalty directly [10].

Table 1. Manifest variables of the constructs.

Construct	Indicators
Likeability	[The company] is a company that I can better identify with than with other companies [The company] is a company that I would more regret not having if it no longer existed than I would other companies I regard [the company] as a likeable company
Competence	[The company] is a top competitor in its market As far as I know, [the company] is recognized worldwide I believe that [the company] performs at a premium level
Quality	The products/services offered by [the company] are of high quality In my opinion [the company] tends to be an innovator, rather than an imitator with respect to [industry] I think that [the company]’s products/services offer good value for money The services [the company] offers are good
	Customer concerns are held in high regards at [the company] [The company] seems to be a reliable partner for customers I regard [the company] as a trustworthy company I have a lot of respect for [the company]
Performance	[The company] is a very well-managed company [The company] is an economically stable company I assess the business risk for [the company] as modest compared with its competitors I think that [the company] has growth potential [The company] has a clear vision about the future of the company
Attractiveness	In my opinion [the company] is successful in attracting high-quality employees I could see myself working at [the company] I like the physical appearance of [the company] (company, buildings, shops, etc.).
CSR	[The company] behaves in a socially conscious way I have the impression that [the company] is forthright in giving information to the public I have the impression that [the company] has a fair attitude towards competitors [the company] is concerned about the preservation of the environment. I have the feeling that [the company] is not only concerned about profits.
Loyalty	I would recommend [company] to friends and relatives If I had to choose again, I would chose [company] as my mobile phone services provider I will remain a customer of [company] in the future
Satisfaction	If you reconsider your experiences with [company] how satisfied are you with [company]?

In reference to the studies by Anderson and Sullivan [2] and Eberl [10], “satisfaction” is operationalized by means of a single item [44], while “loyalty” is measured using three reflective items well known from empirical marketing studies [10]. The mode for measuring latent variables (formative or reflective) was theoretically established in prior research works [48]. A confirmatory tetrad analysis in PLS-PM [18] allows us to empirically test the directionality of the relationships in each measurement model, but does not provide new insights that cast doubt on the theoretically established formative indexes and reflective scales in this study. Figure 1 illustrates the path model under consideration.

To estimate the PLS path model, data were collected from four major service providers in Germany’s mobile communications market. This was done by means of CATI interviews in February 2005. The respondents rated the manifest indicators in the PLS path model on 7-point Likert scales, with higher scores denoting higher variable levels. Satisfaction and loyalty were surveyed in respect of the interviewees’ own service providers. The dataset comprises $N = 344$ subjects representing the following four stakeholder groups: general public ($n = 210, s = 1$), representatives of the media ($n = 34, s = 2$), politics ($n = 50, s = 3$), and the financial community ($n = 50, s = 4$).

Building on these data, the PLS path model estimation was carried out by means of SmartPLS 2.0 [38], the only statistical software application for graphical path modelling with latent variables that employs FIMIX-PLS capabilities. Figure 1 and Table 2 present the PLS-PM results on the aggregate data level **global**.

3.2 Segmentation

Eberl’s [10] analysis reveals that different stakeholder groups (i.e. media, politicians, the financial community, and general public) assess the companies’ reputation and behaviour differently, leading to marketing activities having varying effects on corporate reputation. The results make a strong case for the consideration of *a priori* information to enhance the analysis’s validity.

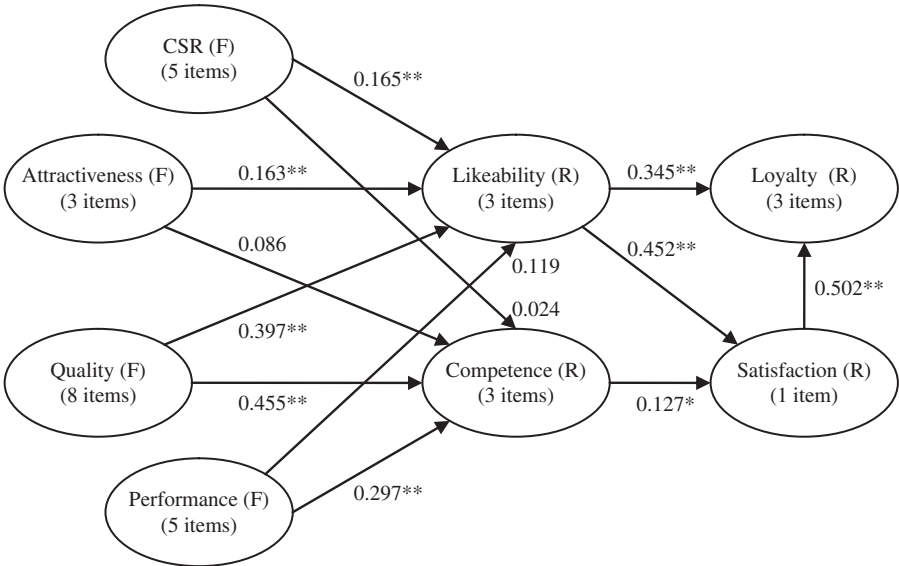


Figure 1. Research model.
Note: Path coefficients stem from the analysis of the global model (Table 2). *Path coefficient significant at 0.10. **Path coefficient significant at 0.05. The index “F” indicates a formative measurement model, “R” a reflective measurement model.

Table 2. Path coefficients and GoF measures (I).

Data analysis strategy	Global	A priori				FIMIX		
		s = 1	s = 2	s = 3	s = 4	s = 1	s = 2	p ₁₂ [mgp]
Quality → competence	0.455**	0.464**	0.771**	0.501**	−0.497**	−0.505**	−0.511**	0.378
Performance → competence	0.297**	0.324**	0.071	0.276**	−0.115*	−0.071	−0.406**	0.042
Attractiveness → competence	0.086	0.030	0.219**	0.060	−0.238**	−0.220**	−0.267*	0.000
CSR → competence	0.024	0.040	−0.202**	0.099	−0.058	−0.407**	−0.293**	0.000
Quality → likeability	0.397**	0.425**	0.146*	0.660**	−0.359**	−0.313**	−0.499**	0.156
Performance → likeability	0.119	0.054	0.665**	0.099	−0.122*	−0.130	−0.287**	0.000
Attractiveness → likeability	0.163**	0.162*	−0.157**	0.078*	−0.352**	−0.192**	−0.114*	0.262
CSR → likeability	0.165**	0.216**	0.164**	0.059	−0.079	−0.205**	−0.075*	0.249
Competence → satisfaction	0.127*	0.174*	0.127*	0.021	−0.030	−0.201**	−0.388**	0.000
Likeability → satisfaction	0.452**	0.436**	0.532**	0.551**	−0.448**	−0.437**	−0.350**	0.302
Satisfaction → loyalty	0.502**	0.513**	0.650**	0.385**	−0.466**	−0.415**	−0.615**	0.035
Likeability → loyalty	0.345**	0.353**	0.147**	0.479**	−0.323**	−0.334**	−0.308**	0.438
R ² _s (competence)	0.631	0.627	0.805	0.789	0.530	0.542	0.803	
R ² (competence)	0.631	0.654				0.715		
R ² _s (likeability)	0.558	0.575	0.570	0.744	0.606	0.254	0.800	
R ² (likeability)	0.558	0.603				0.615		
R ² _s (satisfaction)	0.293	0.318	0.372	0.321	0.186	0.173	0.485	
R ² (satisfaction)	0.293	0.304				0.380		
R ² _s (loyalty)	0.556	0.587	0.558	0.587	0.451	0.387	0.719	
R ² (loyalty)	0.556	0.564				0.607		
GoF _s	0.605	0.604	0.671	0.677	0.561	0.472	0.724	
GoF	0.605	0.615				0.639		
π _s	1	0.610	0.099	0.145	0.145	0.337	0.663	
n _s	344	210	34	50	50	116	228	

Note: p₁₂ [mgp], *p*-value for multi-group comparison test by Henseler [20] for path differences between *s* = 1 and *s* = 2.
*Path coefficient significant at 0.10.
**Path coefficient significant at 0.05.

However, researchers and practitioners never know if they have arrived at an effective segmentation result since heterogeneity may also be unobservable. Consequently, observations cannot be easily divided into subpopulations and the usage of FIMIX-PLS may provide even further differentiated results.

A systematic application of FIMIX-PLS that includes an ex-post analysis requires four key steps [36]. In the first step, the standard PLS-PM algorithm is applied to the aggregate data to

obtain latent variable scores for each observation. These latent variable scores are then used as input for the second step in which FIMIX-PLS is run with a user-specified number of classes. In the third step, a subsequent ex-post analysis aims to identify explanatory variables that lead to a partition similar to the one obtained by FIMIX-PLS. This new partition is then used in the fourth step to calculate local models whose model parameters can be compared by means of PLS-PM multi-group comparison procedures.

Consequently, FIMIX-PLS was used on the data and the procedure was repeated, using consecutive numbers of segments s and 10 replications. The adequate model, i.e. the number of segments, was chosen according to the minimal value of the heuristic CAIC measure, which has been substantiated as working particularly well with FIMIX-PLS [47]. According to the results, the two-segment solution is deemed appropriate. The results of the other information criteria and the EN support this finding (Table 3).

Table 3 also presents the development of FIMIX-PLS segment sizes. The solutions for five and more classes are not admissible. The smallest segment size attains levels of 0.075, which does not provide enough observations to conduct a reasonable segment-specific PLS path analysis. Moreover, from a marketing perspective, segments of this size are relatively unimportant for interpretations. FIMIX-PLS provides probabilities of membership for each observation to fit in any of the two segments. To partition the data on the basis of these results, each observation is assigned to a segment according to its maximum probability of segment membership. The two-segment solution identifies a smaller ($\pi_1 = 0.337$) and a larger segment ($\pi_2 = 0.663$). Subsequently, each segment is analysed separately by applying the standard PLS-PM algorithm to each set of data. The results of this analysis **FIMIX** are presented in the following section.

In the third step, an ex-post analysis is conducted that aims at identifying an explanatory variable for partitioning data in accordance with the FIMIX-PLS results regarding the most appropriate number of classes. This kind of *a priori* segmentation and group-specific estimation of the PLS path model is crucial for updating the measurement model parameters, which are kept constant in the initial FIMIX-PLS algorithm. Hahn *et al.* [19] suggest that, as proposed by Ramaswamy *et al.* [32], an ex-post analysis should be undertaken of a modified form of a segment membership's estimated probabilities. Contingency table testing, logistic regression, or discriminant analyses may be likewise applied to identify variables that can be used to classify additional observations to one of the designed segments. This research applies an exhaustive CHAID analysis as presented by Ringle *et al.* [36] in the context of FIMIX-PLS.

The analysis shows that given the potential explanatory variables in the survey data, the differentiation between business users ($s = 1$) and private users ($s = 2$) is the best fitting segmentation. Based on this result, the data set is partitioned into two segments, which are subsequently used as input for group-specific PLS-PM analyses. The results of this analysis step **FIMIX ex-post** are presented in Section 3.3.

In accordance with Ringle [33] and, later, Esposito Vinzi *et al.* [12], different data analysis strategies were applied to compare the performance of FIMIX-PLS with that of ordinary PLS path

Table 3. FIMIX-PLS evaluation criteria and relative segment sizes.^a

S	Akaike's information criterion (AIC)	Bayesian information criteria (BIC)	Consistent AIC (CAIC)	Normed entropy statistic (EN)	Relative segment sizes π_s			
					$s = 1$	$s = 2$	$s = 3$	$s = 4$
2	3117.81	3244.55	3277.55	0.49	0.34	0.66		
3	3236.06	3428.10	3478.10	0.42	0.40	0.39	0.21	
4	3314.88	3572.21	3639.21	0.46	0.37	0.25	0.22	0.16

^aSection 2 introduces the evaluation criteria in this table.

model analyses in distinct subsamples derived from *a priori* information, or from a clustering procedure:

- Based on the modalities of the variable “stakeholder group”, the same PLS path model was estimated in four distinct subpopulations (**a priori**).
- Using *k*-means clustering – the best known and most widely used non-hierarchical clustering procedure [56] – on the values of the manifest variables, the overall sample was – in keeping with the results of **FIMIX** – partitioned into two subpopulations, each of which was analysed by means of the standard PLS-PM algorithm (**k-means mv**).
- The same procedure was followed as in **k-means mv**, but clustering was performed on the latent variable scores, which had been derived from **global (k-means lv)**.³

A comparison of the FIMIX-PLS results with the group-specific PLS-PM outcomes (obtained by alternative approaches) allows us to assess the approach’s effectiveness and capabilities in an initial application to a complex path model in the field of marketing. The systematic evaluation of PLS-PM results uses the procedure suggested by Henseler *et al.* [21] to apply a set of non-parametric evaluation criteria for PLS-PM as put forward by Chin [5] as well as by Tenenhaus *et al.* [53].

3.3 Evaluation of the measurement models

To ensure that the measures are reliable and valid, item reliabilities, the reliability of the constructs (also referred to as convergent validity) as well as the discriminant validity are evaluated [21]. In PLS-PM, the reliability of reflective items is assessed by examining the loadings of each item with the associated construct. Generally, loadings should approach or exceed 0.70 to be considered for

Table 4. Reliability, validity, and GoF measures for [*k*-means mv] with $S = 3$ and $S = 4$.

Data analysis strategy	<i>k</i> -means mv ($S = 3$)			<i>k</i> -means mv ($S = 4$)			
	$s = 1$	$s = 2$	$s = 3$	$s = 1$	$s = 2$	$s = 3$	$s = 4$
ρ_c (competence)	0.810	0.748	0.743	0.708	0.734	0.789	0.739
ρ_c (likeability)	0.667	0.758	0.842	0.695	0.734	0.497	0.795
ρ_c (loyalty)	0.842	0.819	0.855	0.479	0.874	0.859	0.828
$\varepsilon_{\text{competence}}$	0.088	0.204	0.157	0.165	0.139	0.033	0.082
$\varepsilon_{\text{likeability}}$	0.066	0.494	0.325	0.200	0.509	−0.037	0.332
$\varepsilon_{\text{loyalty}}$	0.197	0.233	0.399	0.198	0.348	0.202	0.443
R_s^2 (competence)	0.526	0.308	0.411	0.335	0.464	0.625	0.554
R^2 (competence)	0.393			0.470			
R_s^2 (likeability)	0.425	0.139	0.250	0.142	0.043	0.415	0.292
R^2 (likeability)	0.240			0.188			
R_s^2 (satisfaction)	0.023	0.058	0.177	0.077	0.026	0.023	0.191
R^2 (satisfaction)	0.102			0.079			
R_s^2 (loyalty)	0.407	0.318	0.201	0.323	0.241	0.420	0.191
R^2 (loyalty)	0.285			0.283			
GoF_s	0.420	0.326	0.378	0.324	0.308	0.410	0.402
GoF	0.366			0.350			
π_s	0.189	0.384	0.427	0.305	0.308	0.157	0.230

Table 5. Reliability, validity, and GoF measures for (*k*-means lv) with *S* = 3 and *S* = 4.

Data analysis strategy	<i>k</i> -means lv (<i>S</i> = 3)			<i>k</i> -means lv (<i>S</i> = 4)			
	<i>s</i> = 1	<i>s</i> = 2	<i>s</i> = 3	<i>s</i> = 1	<i>s</i> = 2	<i>s</i> = 3	<i>s</i> = 4
ρ_c (competence)	0.727	0.758	0.820	0.651	0.684	0.843	0.744
ρ_c (likeability)	0.761	0.828	0.752	0.730	0.820	0.745	0.768
ρ_c (loyalty)	0.771	0.869	0.834	0.843	0.872	0.761	0.715
$\varepsilon_{\text{competence}}$	0.192	0.124	0.136	0.263	0.218	0.050	0.213
$\varepsilon_{\text{likeability}}$	0.467	0.355	0.069	0.453	0.406	0.043	0.452
$\varepsilon_{\text{loyalty}}$	0.407	0.361	0.250	0.358	0.463	0.458	0.214
R_s^2 (competence)	0.327	0.422	0.499	0.324	0.525	0.617	0.328
R^2 (competence)	0.399			0.410			
R_s^2 (likeability)	0.132	0.225	0.447	0.050	0.230	0.497	0.232
R^2 (likeability)	0.230			0.195			
R_s^2 (satisfaction)	0.086	0.185	0.002	0.059	0.123	0.011	0.069
R^2 (satisfaction)	0.113			0.071			
R_s^2 (loyalty)	0.251	0.255	0.320	0.221	0.179	0.124	0.245
R^2 (loyalty)	0.265			0.205			
GoF _{<i>S</i>}	0.318	0.391	0.420	0.279	0.368	0.422	0.337
GoF	0.368			0.333			
π_s	0.387	0.430	0.183	0.381	0.244	0.122	0.253

further analysis [41]. Other authors deem values of above 0.40 or 0.50 as appropriate [23]. This reliability analysis applies predominantly to reflective measurement models, as these covariance-based procedures are not applicable to the constructs measured by formative indicators. In contrast to single regression coefficients (loadings) in the reflective case, interrelations between manifest variables and constructs represent multiple regression coefficients (weights) in the formative case. To evaluate formative measurement models, Chin [5] suggests a threshold value of 0.10 for formative measures, with the weights being tested for significance by means of the bootstrapping procedure.

Across all models, the loadings of the reflective set are uniformly high – above 0.80. Among the formative measures, weights are significant at 0.05 and lie well above the suggested threshold of 0.10 in almost all cases. Few exceptions can be observed with regard to measures of the constructs “quality” and “CSR”. However, these deviations occur in all models and are thus independent of the associated analysis context. Consequently, in a marginal number of relationships, low weights are not considered problematic for the present analysis.

The reliability of constructs using reflective indicators is assessed by means of composite reliability ρ_c [57]. Traditionally, Cronbach’s α [7] is used for reliability analyses. This criterion systematically provides better outcomes with an increasing number of indicators and assumes the indicators’ τ -equivalency. Consequently, the composite reliability measure ρ_c [57], which does not require indicators to be equally reliable, has become the primary criterion of choice for evaluating reflective measurement models in PLS-PM [21]. In this PLS-PM application, the ρ_c on the aggregate data level and both FIMIX-PLS analyses of the reflectively measured endogenous latent variables (not including the single-indicator “satisfaction” construct) range between 0.81 and 0.91 (Table 6). These values as well as the results of every single outer loading do not significantly differ when the group-specific FIMIX-PLS results are compared, or when contrasting each of these PLS-PM computations with the outcomes on the aggregate data level.

Table 6. Discriminant validity and reliability measures.

Measure	Global	<i>A priori</i>				FIMIX		FIMIX ex post		<i>k</i> -means mv		<i>k</i> -means lv	
		<i>s</i> = 1		<i>s</i> = 2	<i>s</i> = 3	<i>s</i> = 4	<i>s</i> = 1	<i>s</i> = 2	<i>s</i> = 1	<i>s</i> = 2	<i>s</i> = 1	<i>s</i> = 2	
ρ_c (competence)	0.869	0.857	0.925	0.885	0.843	0.815	0.895	0.884	0.865	0.817	0.789	0.816	0.778
ρ_c (likeability)	0.899	0.886	0.904	0.916	0.914	0.883	0.902	0.907	0.895	0.802	0.827	0.805	0.830
ρ_c (loyalty)	0.894	0.911	0.894	0.844	0.872	0.905	0.880	0.880	0.899	0.847	0.857	0.843	0.854
$\varepsilon_{\text{competence}}$	0.064	0.057	0.029	−0.024	0.100	0.133	0.038	0.531	0.386	0.158	0.128	0.188	0.123
$\varepsilon_{\text{likeability}}$	0.150	0.129	0.141	0.027	0.178	0.359	0.003	0.440	0.319	0.309	0.397	0.254	0.414
$\varepsilon_{\text{loyalty}}$	0.173	0.173	0.121	0.106	0.228	0.333	0.028	0.616	0.449	0.180	0.375	0.201	0.375
π_s	1	0.610	0.099	0.145	0.145	0.337	0.663	0.267	0.733	0.462	0.538	0.445	0.555
<i>N</i>	344	210	34	50	50	116	228	92	252	159	185	153	191

Note: ρ_c , composite reliability; ε , measure for criterion by Fornell and Larcker [15]; π_s , relative size of segment *s*; *n*, number of observations.

In the next step, the discriminant validity of the endogenous constructs is evaluated. To assess the extent to which the measures do not correlate with other constructs (from which they are supposed to differ), two approaches are applied. First, the indicators' cross-loadings are examined, which reveals that no indicator loads higher on the opposing endogenous constructs. Second, the Fornell and Larcker criterion [15] is applied, which compares the square root of each endogenous construct's average variance extracted (AVE) and its bivariate correlations with all opposing endogenous constructs [17]. This can be expressed as

$$\varepsilon_m \sqrt{\text{AVE}_m} - \max_{l=1, \dots, L} (\rho_{ml}) \quad \forall m \neq l \quad (l = 1, \dots, L, m = 1, \dots, M), \quad (3)$$

where AVE_m is the AVE for the endogenous construct m and $\max(\rho_{ml})$ denotes the maximum value of all correlations between this construct and an opposing endogenous construct l . For adequate discriminant validity, ε_m should be significantly greater than zero [23]. As with ρ_c , the Fornell and Larcker criterion is not applicable to constructs measured with a single item.

The results in Table 6 show that with a single exception in **a priori**, the square root of AVE is greater than the variance shared between each construct and its opposing constructs. However, with regard to the larger segment in **FIMIX**, the values for ε_m range only slightly above zero, indicating that the constructs and their measures cannot be adequately discriminated. Strictly speaking, it is inappropriate to view the constructs of this segment as distinct and separate theoretical entities. Based on the set-up of the present study, it is not possible to evaluate whether this is a methodological or data-related issue. Even though the criterion applied by Fornell and Larcker [15] is very conservative, these results need to be further elaborated in subsequent simulation studies.

Finally, when conducting a PLS-PM multi-group analysis to compare segment-specific estimates, the "establishment of measurement invariance across groups is a logical prerequisite to conducting substantive cross-group comparisons" [55]. The results allow one to ascertain that the measurement parameters (factor loadings, measurement errors, etc.) are the same across groups and that the same constructs are measured in all groups. Researchers can thus be certain that they compare the same kind of occurrences when interpreting significant differences in path modelling results obtained from FIMIX-PLS and subsequent PLS-PM multi-group analyses. However, there are as yet no appropriate methods for this analysis in respect of PLS-PM, which should be the subject of future research.⁴ An appropriate means of testing measurement model invariance in PLS-PM may build on bootstrapping or permutation test-based PLS-PM multi-group analysis results [9,26] to answer the following four questions [51]: (1) Are the measurement parameters (factor loadings, measurement errors, etc.) the same across all groups?; (2) Are there pronounced response biases in a particular group?; (3) Can one unambiguously interpret observed mean differences as latent mean differences?; and (4) Is the same construct measured in all groups? Future research must present an appropriate procedure and an adequate test to evaluate measurement model invariance, not only in respect of reflective measurement models, but also regarding formative measurement models in PLS-PM. This future contribution will be of central relevance when conducting the PLS-PM multi-group analysis in the last step of a systematic application of the FIMIX-PLS approach.

3.4 Evaluation of the inner model

Unlike the parametric CSA analysis that minimizes the difference between observed and reproduced covariance when estimating model parameters, the PLS-PM algorithm does not optimize any global scalar function and, hence, does not offer comparable global GoF criteria. The central evaluation criteria for the inner PLS path model are the R^2 values of the endogenous latent variables [22]. In contrast, the GoF, as suggested by Tenenhaus *et al.* [52], allows both reflective measurement model evaluations and inner model ones to be incorporated within a single criterion.

To validate the PLS path model globally, the authors proposed an index (normed between zero and one) as the geometric mean of the average communality and the average R^2 [53]:

$$\text{GoF} = \sqrt{\text{communality} \cdot \overline{R^2}} \quad (4)$$

Another central criterion for the evaluation of the inner model is the analysis of the interrelations between the constructs. For the evaluation of path coefficients, t -values were calculated using the bootstrapping procedure with 2000 subsamples and a number of cases per subsample, which equals to the number of observations in the segment under consideration. The segment-specific path coefficients were tested for significant differences by using a multi-group comparison procedure by Henseler [20]. The newly proposed approach bootstraps a single parameter of interest and compares all possible combinations of bootstrap parameters between the segments. This kind of comparison allows the probability to be determined that the path coefficient in a segment under consideration is smaller than or equal to its counterpart in the other segment. Henseler's [20] procedure was preferred, as it became apparent in the computation that this PLS-PM typological multi-group analysis approach produces more conservative results, i.e. higher p -values compared with the parametric testing procedure in PLS-PM [26]. Tables 2 and 7 provide an overview of the path coefficients in the models, the mixing proportions of each segment π_s , the GoF measures, and the R^2 values of all endogenous latent constructs. These tables also indicate the significance level of each path coefficient and the significance of segment-specific path differences.

With regard to the global model, the GoF value (0.605) as well as the R^2 values of the constructs "competence", "likeability", and "loyalty" is very acceptable. As indicated by Eberl [10], the rather low value in respect of "satisfaction" is not surprising since intangible assets, like corporate reputation, are only one of the numerous customer satisfaction determinants. In contrast, both data analysis strategies that apply k -means clustering (Table 7) show considerably lower values for these measures, thus disclosing an unacceptable model fit. The multi-group comparison test for these data analysis strategies reveals that, overall, only five path coefficients differ significantly at 0.10 between the two segments. This allows only very limited conclusions, especially since about half of the paths are not significant at 0.05. The results of **k -means mv** and **k -means lv** underline notions in previous research [12,24,33] that *a priori* clustering approaches are inappropriate, as they do not allow the statistical model, i.e. the path relationships established by the researcher, to be considered.

When comparing the global model with the results derived from the separate analysis of the stakeholder groups (**a priori**; Table 2), one finds that the relative importance of "quality", "performance", "attractiveness", and "CSR" differs quite substantially within the four subsamples, with some relationships showing sign changes. In the media subsample ($s = 2$), specifically, most path estimators are significant and very distinct from the paths in the other groups [10]. However, in the other stakeholder groups, only about half of the paths are significant at 0.05. Nevertheless, the overall values for R^2 and GoF for **a priori** range slightly above the PLS-PM results for **global**. Compared with this outcome, the FIMIX-PLS results (Table 2) are much more persuasive. All endogenous constructs have increased overall R^2 values, ranging between 9% ("loyalty") and 30% ("satisfaction") higher than in *global*. In comparison with *a priori*, the overall R^2 values range between 2% ("likeability") and 25% ("satisfaction") higher. However, this is specifically due to the second segment's results in which the R^2 values lie considerably above the other models' results. Conversely, the first segment exhibits rather low values compared with the other data analysis strategies, indicating a poor model fit. This holds specifically for the construct "likeability" whose R^2 values range even below those of **k -means mv** and **k -means lv**. With a value of 0.639, the overall GoF is highly satisfactory compared with **global** and **a priori** results.

Table 7. Path coefficients and GoF measures (II).

Data analysis strategy	FIMIX ex-post			k-means mv			k-means lv		
	p_{12} [mgp]		p_{12} [mgp]	p_{12} [mgp]		p_{12} [mgp]	p_{12} [mgp]		p_{12} [mgp]
	$s = 1$	$s = 2$		$s = 1$	$s = 2$		$s = 1$	$s = 2$	
Quality $\rightarrow$ competence	0.602**	0.440**	0.077	0.397**	0.421**	0.429	0.372**	0.415**	0.415
Performance $\rightarrow$ competence	0.097	0.340**	0.020	0.253**	0.277**	0.518	0.266**	0.270**	0.503
Attractiveness $\rightarrow$ competence	0.213**	0.041	0.049	0.058	0.118*	0.531	0.040	0.127*	0.330
CSR $\rightarrow$ competence	-0.061	0.058	0.130	0.123*	0.006	0.115	0.113	-0.020	0.022
Quality $\rightarrow$ likeability	0.178**	0.426**	0.029	0.296**	0.171*	0.443	0.370**	0.160*	0.199
Performance $\rightarrow$ likeability	0.345**	0.082	0.013	-0.005	0.222*	0.000	0.025	0.184*	0.355
Attractiveness $\rightarrow$ likeability	0.194**	0.137**	0.358	0.136	0.141	0.374	0.057	0.147*	0.369
CSR $\rightarrow$ likeability	0.114**	0.230**	0.132	0.264**	0.072	0.167	0.229**	0.095	0.331
Competence $\rightarrow$ satisfaction	0.054	0.142*	0.273	-0.037	0.171**	0.000	-0.100	0.182**	0.145
Likeability $\rightarrow$ satisfaction	0.476**	0.446**	0.386	0.247**	0.301**	0.415	0.245**	0.279**	0.267
Satisfaction $\rightarrow$ loyalty	0.543**	0.492**	0.251	0.569**	0.374**	0.046	0.548**	0.372**	0.084
Likeability $\rightarrow$ loyalty	0.250**	0.378**	0.050	0.247**	0.192**	0.332	0.258**	0.203**	0.452
π_s	0.244	0.756		0.462	0.538		0.445	0.555	
R^2_s (competence)	0.655	0.659		0.443	0.463		0.409	0.448	
R^2_s (competence)	0.658			0.454			0.431		
R^2_s (likeability)	0.500	0.609		0.273	0.193		0.303	0.195	
R^2_s (likeability)	0.580			0.230			0.243		
R^2_s (satisfaction)	0.257	0.304		0.055	0.152		0.051	0.139	
R^2_s (satisfaction)	0.291			0.107			0.100		
R^2_s (loyalty)	0.495	0.585		0.452	0.228		0.426	0.230	
R^2_s (loyalty)	0.561			0.331			0.317		
GoF _s	0.595	0.619		0.374	0.389		0.418	0.383	
GoF	0.613			0.386			0.399		
π_s	0.267	0.733		0.462	0.538		0.445	0.555	
n_s	92	252		159	185		153	191	

Note: p_{12} [mgp], p -value for multi-group comparison test by Henseler [20] for path differences between $s = 1$ and $s = 2$.

*Path coefficient significant at 0.10.

**Path coefficient significant at 0.05.

801 Despite this poor result of the first segment's analysis, almost all path coefficients have been
proven to be significant at 0.05 in **FIMIX**. For example, the global model suggests that "attractive-
ness" is not a significant antecedent of corporate reputation's cognitive dimension. This finding
implies that in the mobile communications market, investments in a favourable assessment of
the company's attractiveness do not pay off in terms of an increase in perceived competence.
806 In **FIMIX** however, this path relationship differs substantially between the two segments and is
significant at 0.05 and 0.10, respectively. In fact, it is clear that the segment-specific relation-
ship is balanced in the global model. Likewise, both segments differ considerably with regard to
the antecedents of customer satisfaction, as it is negatively influenced by perceived competence
in the smaller segment and positively in the larger segment. A multi-group comparison reveals
811 that about half the relationships differ significantly between two segments, which is considerably
higher than in the other analysis strategies. This number could even be increased by applying a
less conservative PLS-PM multi-group comparison procedure [9,26].

The results from the FIMIX-PLS ex-post analysis (**FIMIX ex-post**; Table 7) also turn out to be
very satisfactory. Most path coefficients are significant at 0.05 and differ considerably between
816 the two segments. Overall, the R^2 values range higher than in **global** and the explanatory power
is better balanced between the two segments compared with that of **FIMIX**. Nevertheless, the
reputation drivers' influence on "likeability" and "competence" differs considerably between the
two segments. Specifically, the four path relationships between the two formative exogenous
latent variables, "quality" and "performance", and the two reflective endogenous latent variables,
821 "competence" and "likeability", vary significantly between business and private users. **FIMIX ex-
post** analysis results provide the outer weights of each formative measurement model. While the
weights in the formative latent variable "quality" do not vary considerably between the segments,
the formative construct "performance" shows significant differences in the outer weights when a
PLS-PM multi-group analysis is undertaken. According to these results, FIMIX-PLS is capable of
826 uncovering heterogeneity. However, in comparison to the media ($s = 2$) and politicians subsample
($s = 3$), the ex-post analysis results lag behind with regard to the R^2 values for "competence"
and "satisfaction".

Furthermore, when comparing the initial FIMIX-PLS with the ex-post analysis results, we
observe that, as expected (Section 2), not all segment-specific path coefficients can be adequately
831 reproduced by means of explanatory variables. Whereas, for example, paths from "performance"
to "competence" or "attractiveness" to "likeability" lie on comparable levels in both analyses,
this is not the case with respect to those related to the satisfaction construct. This shows that the
explanatory variables in the dataset have only limited potential for an ex-post analysis that mirrors
initial FIMIX-PLS results appropriately in every respect. This result underlines that a FIMIX-
836 PLS ex-post analysis is highly dependent on the availability of suitable explanatory variables
to form data groups which have PLS-PM estimates and segment sizes similar to that of the
FIMIX-PLS group-specific outcomes. With the exception of PATHMOX, this notion also holds for
alternative PLS segmentation approaches such as PLS-GAS and REBUS-PLS. Even though these
approaches do not rely on an ex-post analysis to dynamically update the measurement models –
841 as with FIMIX-PLS – the resulting segments are unobservable and, thus, are of little use in real-
world applications. Managers should be able to identify customers in each segment by means
of easily measured variables. This is a prerequisite for assessing the substantiality, accessibility,
stability, responsiveness, and actionability of any segmentation result [56]. Consequently, an ex-
post analysis must likewise be carried out when using alternative PLS-PM clustering procedures.
846 Hence, it is of utmost importance to collect a high number of potential explanatory variables when
conduction surveys to ensure the effectiveness of the segmentation strategy in applied statistics.

Overall, the results demonstrate that an aggregate analysis of reputation data is seriously mis-
leading and results in flawed inferences and management implications. By incorporating *a priori*
information, the model fit could be increased and segment-specific results could enable marketing

managers to meet relevant target groups' needs more accurately. The defined model, as well as the implied relationships between the latent variables, is taken into account if FIMIX-PLS is applied, resulting in an increased model fit and differentiated paths in the inner model. Results based on the FIMIX-PLS ex-post analysis could not meet this GoF but still proved value by increasing the accuracy of the results. Finally, both the data analysis strategies that apply *k*-means clustering proved to be unsuitable for forming segments with distinctive path model estimates in the inner model. These results thus confirm previous findings in the context of CSA [24,58] and underpin the results of Ringle's study [33] in a more complex model set-up.

4. Discussion and conclusion

Unobserved heterogeneity and measurement errors are endemic problems in social sciences. Jedidi *et al.* [24] have addressed these problems in respect of structural equation modelling. Hahn *et al.* [19] have further developed the finite-mixture SEM methodology for PLS-PM. Besides PLS-GAS [34,37] and REBUS-PLS [13], FIMIX-PLS can currently be regarded as a key approach with which to capture heterogeneity in PLS-PM, despite some limitations [42]. The FIMIX-PLS algorithm has been integrated into the easy-to-use statistical software application SmartPLS [38], which can be used in various fields of marketing.

This article answers Esposito Vinzi *et al.*'s [12] call for further research by evaluating the effectiveness of the FIMIX-PLS approach in a complex model set-up with empirical data from the field of marketing. The greater number of constructs and manifest variables, as well as the integration of both formative and reflective measurement models, mirror actual applications in the consumer research community more accurately. This is necessary to illustrate the applicability of FIMIX-PLS from a practical point of view. Furthermore, the application of FIMIX-PLS further extends Eberl's [10] recent study of corporate-level marketing activities' effects on corporate reputation in the German mobile communications industry. Moreover, the segmentation results are compared with the use of *a priori* information and the clustering procedure *k*-means on the manifest and latent variable scores level.

While sequential procedures of *k*-means clustering have been proven to be inappropriate, FIMIX-PLS captures unobserved heterogeneity adequately. The group-specific PLS-PM results enhance the overall explanatory power of the inner model, which is mirrored in increased R^2 and GoF values. The present study shows that this improvement does not come at the expense of the scales' reduced reliability and validity in the measurement models. On the whole, the evaluation of validity and reliability produced highly satisfactory results. Moreover, significant differences in the segment-specific inner model path coefficients allow for more precise interpretations and effective marketing strategies.

This article's main contribution to the body of knowledge on clustering data in PLS-PM is twofold. First, the analysis shows that the PLS-PM results and their interpretation can be misleading if (unobserved) heterogeneity affects inner model estimates. Previous publications on PLS-PM largely ignored this critical issue [45]. Second, the article illustrates that FIMIX-PLS uncovers unobserved heterogeneity reliably and offers a solution for this problem by clustering observations into the best fitting number of classes with distinct inner model path estimates. FIMIX-PLS is generally applicable to reflectively measured latent variables as well as to formatively measured ones. In the latter case, heterogeneity in the inner model may have its source in homogenous or heterogeneous outer weight relationships. Based on **FIMIX** results, which uncover unobserved heterogeneity in the inner model in respect of the latent variable scores, the **FIMIX ex-post** outcomes do not only replicate these results in respect of private and business users, but also uncover, for example, the homogenous weights pattern of the formatively measured exogenous latent variable "quality" as well as the significantly differing outer weights of the exogenous latent variable "performance".

The results underline that PLS-PM applications should exploit this approach to response-based market segmentation by identifying certain groups of customers – as long as unobserved moderating factors explain consumer heterogeneity within the inner model relationships. Consequently, researchers and practitioners gain the confidence they need to decide whether or not unobserved heterogeneity affects path modelling outcomes. If the outcomes on the aggregate data level exhibit significantly different group-specific PLS-PM estimates, the systematic FIMIX-PLS approach to path modelling provides further differentiated and more effective analytical results and conclusions. We expect these conditions to hold true in many marketing-related PLS-PM applications.

While prior research addressed FIMIX-PLS issues regarding non-normal data [12] and the applicability of this methodology when latent variables in the path model have formative measurement models [39], this research addresses the potential of FIMIX-PLS to capture heterogeneity in complex path model constellations. Nevertheless, the results of this study are not without limitations. As described in detail by Ringle [33], future research must address critical issues linked to the FIMIX-PLS methodology and the EM algorithm, such as convergence in local optimum solutions [27,54]. Methodological advances must address the remaining significant FIMIX-PLS issues, while extensive simulation studies with varying factors (e.g. sample size, number of indicators, distributional characteristics of the data, relative segment size, number of segments, model complexity) must substantiate the findings of this study as well as those of previous publications. As indicated by this study's results, special attention should be paid to the evaluation of the segment-specific constructs' discriminant validity. Moreover, future research should look for better ways to profile segments, using observable variables in the course of an ex-post analysis of the FIMIX-PLS results. The analysis results clearly show the necessity to identify as many potential explanatory variables as possible to allow for a successful reproduction of the initial FIMIX-PLS partition.

From a formal, statistical point of view, great progress has recently been made in the field of segmentation in PLS-PM. The presented approaches utilize, for example, finite mixture, fuzzy regression, and genetic algorithm concepts, and differ substantially in terms of the covered types of heterogeneity, the distributional assumptions, and interpretability of the resulting segments. As soon as these methods are commonly available in software tools, subsequent simulation studies should also include these approaches. An evaluation and comparison of the results must aim at determining the best PLS-PM segmentation method that researchers and practitioners can apply in their specific analysis.

Notes

1. This critical issue is specifically addressed at the end of this section and in Section 4.
2. The evaluation results of the EN and information criteria are often inconsistent. Consequently, criteria have been proposed that combine both concepts. Biernacki and Govaert [3] present the classification likelihood criterion that integrates a penalty term by measuring the fuzziness of the partition, thus providing a compromise between the GoF and the mixture model's ability to provide a meaningful classification. Similarly, the integrated completed likelihood-BIC [4] penalizes the BIC criterion with the estimated entropy in order to account for overlapping segments.
3. In addition to the two-segment analysis using k -means, models were also computed for a three- and four-segment solution. These analyses did not yield differing results. Reliability, validity, and GoF results are provided in Tables 4 and 5.

References

- [1] S. Albers, *PLS and success factor studies in marketing*, in *Handbook of Partial Least Squares: Concepts, Methods and Applications in Marketing and Related Fields*, V. Esposito Vinzi, W.W. Chin, J. Henseler, and H.-F. Wang, eds., Springer, Berlin, 2009, in print.

- 951 [2] E.W. Anderson and M.W. Sullivan, *The antecedents and consequences of customer satisfaction for firms*, Mark. Sci. 12 (1993), pp. 125–143.
- [3] C. Biernacki and G. Govaert, *Using the classification likelihood to choose the number of clusters*, Comput. Sci. Stat. 29 (1997), pp. 451–457.
- [4] C. Biernacki, G. Celeux, and G. Govaert, *Assessing a mixture model for clustering with the integrated completed likelihood*, IEEE Trans. Pattern Anal. Mach. Intell. 22 (2000), pp. 719–725.
- 956 [5] W.W. Chin, *The partial least squares approach for structural equation modeling*, in *Modern Methods for Business Research*, G.A. Marcoulides, ed., Lawrence Erlbaum Associates, London, 1998, pp. 295–336.
- [6] W.W. Chin, *A permutation procedure for multi-group comparison of PLS models*, in *PLS and Related Methods – Proceedings of the International Symposium PLS'03*, M.J. Vilarés, M. Tenenhaus, P.S. Coelho, V. Esposito Vinzi, and A. Morineau, eds., Decisia, Lisbon, 2003, pp. 33–43.
- [7] L.J. Cronbach, *Coefficient alpha and the internal structure of tests*, Psychometrika 16 (1951), pp. 297–334.
- 961 [8] A. Diamantopoulos and H.M. Winklhofer, *Index construction with formative indicators: An alternative to scale development*, J. Mark. Res. 38 (2001), pp. 269–277.
- [9] J. Dibbern and W.W. Chin, *Multi-group comparison: Testing a PLS model on the sourcing of application software services across Germany and the USA using a permutation based algorithm*, in *Handbuch PLS-Pfadmodellierung: Methode, Anwendung, Praxisbeispiele*, F.W. Bliemel, A. Eggert, G. Fassott, and J. Henseler, eds., Schäffer-Poeschel, Stuttgart, 2005, pp. 135–160.
- 966 [10] M. Eberl, *An application of PLS in multi-group analysis: The need for differentiated corporate-level marketing in the mobile communications industry*, in *Handbook of Partial Least Squares: Concepts, Methods and Applications in Marketing and Related Fields*, V. Esposito Vinzi, W.W. Chin, J. Henseler, and H.-F. Wang, eds., Springer, Berlin, 2009, in print.
- Q5 [11] M. Eberl and M. Schwaiger, *Corporate reputation: Disentangling the effects on financial performance*, Eur. J. Mark. 39 (2005), pp. 838–854.
- 971 [12] V. Esposito Vinzi, C.M. Ringle, S. Squillacciotti, and L. Trinchera, *Capturing and treating heterogeneity by response based segmentation in PLS path modeling. A comparison of alternative methods by computational experiments*, DR-07019, Essec Research Center, 2007.
- Q6 [13] V. Esposito Vinzi, L. Trinchera, S. Squillacciotti, and M. Tenenhaus, *REBUS-PLS: A response-based procedure for detecting unit segments in PLS path modeling*, Appl. Stoch. Models Bus. Ind. 24 (2008), pp. 439–458.
- 976 [14] C. Fornell and F.L. Bookstein, *Two structural equation models: LISREL and PLS applied to consumer exit-voice theory*, J. Mark. Res. 19 (1982), pp. 440–452.
- [15] C. Fornell and D.F. Larcker, *Evaluating structural equation models with unobservable variables and measurement error*, J. Mark. Res. 18 (1981), pp. 39–50.
- [16] S. Frühwirth-Schnatter, *Finite Mixture and Markov Switching Models*, Springer, New York, 2006.
- [17] Y. Grégoire and R.J. Fisher, *The effects of relationship quality on customer retaliation*, Mark. Lett. 17 (2006), pp. 31–46.
- 981 [18] S.P. Gudergan, C.M. Ringle, S. Wende, and A. Will, *Confirmatory tetrad analysis in PLS path modeling*, J. Bus. Res. 61 (2008), pp. 1238–1249.
- [19] C.H. Hahn, M.D. Johnson, A. Herrmann, and F. Huber, *Capturing customer heterogeneity using a finite mixture PLS approach*, Schmalenbach Bus. Rev. 54 (2002), pp. 243–269.
- [20] J. Henseler, *A new and simple approach to multi-group analysis in partial least squares path modeling*, in *Proceedings of the PLS'07 International Symposium on PLS and Related Methods – Causalities Explored by Indirect Observation*, H. Martens, T. Næs, and M. Martens, eds., Matforsk, Ås, Norway, 2007, pp. 104–107.
- 986 [21] J. Henseler, C.M. Ringle, and R.R. Sinkovics, *The use of partial least squares path modeling in international marketing*, in *Advances in International Marketing*, Vol. 20 (Special Volume on New Challenges to International Marketing), R.R. Sinkovics and P.N. Ghauri, eds., 2009, pp. 277–320.
- [22] M. Höck and C.M. Ringle, *Strategic networks in the software industry: An empirical analysis of the value continuum*, Proceedings of the IFSAM VIIIth World Congress, Berlin, 2006.
- 991 [23] J. Hulland, *Use of partial least squares (PLS) in strategic management research: A review of four recent studies*, Strateg. Manage. J. 20 (1999), pp. 195–204.
- [24] K. Jedidi, H.S. Jagpal, and W.S. DeSarbo, *Finite-mixture structural equation models for response-based segmentation and unobserved heterogeneity*, Mark. Sci. 16 (1997), pp. 39–59.
- [25] W.A. Kamakura and G.J. Russell, *A probabilistic choice model for market segmentation and elasticity structure*, J. Mark. Res. 26 (1989), pp. 379–390.
- 996 [26] M. Keil, B.C. Tan, K.-K. Wei, T. Saarinen, V. Tuunainen, and A. Wassenaar, *A cross-cultural study on escalation of commitment behavior in software projects*, Manage. Inf. Syst. Quart. 24 (2000), pp. 299–325.
- [27] G. McLachlan and T. Krishnan, *The EM Algorithm and Extensions*, Wiley & Sons, New York, 1997.
- [28] G.J. McLachlan and D. Peel, *Finite Mixture Models*, Wiley & Sons, New York, 2000.
- [29] L.K. Muthén and B.O. Muthén, *Mplus User's Guide*, 4th ed., Muthén & Muthén, Los Angeles, 1998.

- 1001 [30] M.-S. Oh and A. Raftery, *Model-based clustering with dissimilarities: A Bayesian approach*, Tech. Rep. 441, Department of Statistics, University of Washington, 2003. Q6
- [31] F. Palumbo, R. Romano, and V. Esposito Vinzi, *Fuzzy PLS path modelling: A new tool for handling sensory data*, in *Data Analysis, Machine Learning, and Applications, Proceedings of the 31st Annual Conference of the German Classification Society*, C. Preisach, H. Burkhardt, L. Schmidt-Thieme, and R. Decker, eds., Springer, Berlin, 2008, pp. 689–696.
- 1006 [32] V. Ramaswamy, W.S. DeSarbo, D.J. Reibstein, and W.T. Robinson, *An empirical pooling approach for estimating mix elasticities with PIMS data*, *Mark. Sci.* 12 (1993), pp. 103–124.
- [33] C.M. Ringle, *Segmentation for path models and unobserved heterogeneity: The finite mixture partial least squares approach*, Research Papers on Marketing and Retailing 35, University of Hamburg, 2006. Q6
- [34] C.M. Ringle and R. Schlittgen, *A genetic segmentation approach for uncovering and separating groups of data in PLS path modeling*, in *Proceedings of the PLS'07 International Symposium on PLS and Related Methods – Causalities Explored by Indirect Observation*, H. Martens, T. Næs, and M. Martens, eds., Matforsk, Ås, Norway, 2007, pp. 75–78.
- 1011 [35] C.M. Ringle, S. Wende, and A. Will, *Customer segmentation with FIMIX-PLS*, in *PLS and Related Methods – Proceedings of the PLS'05 International Symposium*, T. Aluja, J. Casanovas, V. Esposito Vinzi, A. Morrineau, and M. Tenenhaus, eds., Decisia, Paris, 2005, pp. 507–514.
- [36] C.M. Ringle, M. Sarstedt, and R. Schlittgen, *Finite mixture and genetic algorithm segmentation in partial least squares path modeling: Identification of multiple segments in a complex path model*, in *Advances in Data Analysis, Data Handling and Business Intelligence*, A. Fink, B. Lausen, W. Seidel, and A. Ultsch, eds., Springer, Heidelberg, Berlin, 2009, in print. Q5
- 1016 [37] C.M. Ringle, M. Sarstedt, and E.A. Mooi, *Response-based segmentation using finite mixture partial least squares: Theoretical foundations and an application to American customer satisfaction index data*, *Annals of Information Systems*, in print. Q5
- 1021 [38] C.M. Ringle, S. Wende, and A. Will, *SmartPLS 2.0 (M3) beta*, Hamburg; software available at <http://www.smartpls.de>.
- [39] C.M. Ringle, S. Wende, and A. Will, *The finite mixture partial least squares approach: Methodology and application*, in *Handbook of Partial Least Squares: Concepts, Methods and Applications in Marketing and Related Fields*, V. Esposito Vinzi, W.W. Chin, J. Henseler, and H.-F. Wang, eds., Springer, Berlin, 2009, in print. Q5
- [40] G. Sánchez and T. Aluja, *PATHMOX: A PLS-PM segmentation algorithm*, in *Proceedings of the IASC Symposium on Knowledge Extraction by Modelling*, 2006 (September), provisional chapter (2006). Q6
- 1026 [41] M. Sarkar, R. Enchambadi, S.T. Cavusgil, and P.S. Aulakh, *The influence of complementarity, compatibility, and relationship capital on alliance performance*, *J. Acad. Mark. Sci.* 29 (2001), pp. 358–373.
- [42] M. Sarstedt, *A review of recent approaches for capturing heterogeneity in partial least squares path modelling*, *J. Model. Manage.* 3 (2008), pp. 140–161.
- [43] M. Sarstedt, *Market segmentation with mixture regression models: Understanding measures that guide model selection*, *J. Target. Meas. Anal. Mark.* 16 (2008), pp. 228–246.
- 1031 [44] M. Sarstedt and P. Wilczynski, *More for less? A comparison of single-item and multi-item measures*, *Bus. Adm. Rev.* 69 (2009), forthcoming. Q5
- [45] M. Sarstedt, C.M. Ringle, and M. Schwaiger, *Extending Festge and Schwaiger's model of customer satisfaction with industrial goods to account for unobserved heterogeneity*, in *Proceedings of the 2009 Winter Educator's Conference of the American Marketing Academy (AMA)*, K. Reynolds and J.C. White, eds., Baltimore, MD, pp. 241–244. Q7
- 1036 [46] M. Sarstedt, C.M. Ringle, M.P. Schloderer, and M. Schwaiger, *Accounting for unobserved heterogeneity in the analysis of antecedents and consequences of corporate reputation: An application of FIMIX-PLS*, *Proceedings of the 37th Annual Conference of the European Marketing Academy (EMAC)*, Brighton, UK.
- [47] M. Sarstedt, M. Schwaiger, and C.M. Ringle, *Determining the number of segments in FIMIX-PLS*, *Proceedings of the 2009 Annual Conference of the Academy of Marketing Science (AMS)*, Baltimore, USA, forthcoming. Q5
- [48] M. Schwaiger, *Components and parameters of corporate reputation – an empirical study*, *Schmalenbach Bus. Rev.* 56 (2004), pp. 46–71.
- 1041 [49] S. Squillacciotti, *Prediction oriented classification in PLS path modeling*, in *Handbook of Partial Least Squares: Concepts, Methods and Applications in Marketing and Related Fields*, V. Esposito Vinzi, W.W. Chin, J. Henseler, and H.-F. Wang, eds., Springer, Berlin, 2009, in print. Q5
- [50] J.-B. Steenkamp and H. Baumgartner, *Assessing measurement invariance in cross national consumer research*, *J. Consum. Res.* 25 (1998), pp. 78–107.
- 1046 [51] H. Steinmetz, P. Schmidt, A. Tina-Booh, S. Wiecezorek, and S.H. Schwartz, *Testing measurement invariance using multigroup CFA: Differences between educational groups in human values measurement*, *Qual. Quantity* 43 (2009), forthcoming. Q5
- [52] M. Tenenhaus, S. Amato, and V. Esposito Vinzi, *A global goodness-of-fit index for PLS structural equation modelling*, *Proceedings of the XLII Società Italiana di Statistica (SIS) Scientific Meeting*, CLEUP, Padova, Italy, 2004, pp. 739–742.

- 1051 [53] M. Tenenhaus, V. Esposito Vinzi, Y.-M. Chatelin, and C. Lauro, *PLS path modeling*, Comput. Stat. Data Anal. 48 (2005), pp. 159–205.
- [54] D.M. Titterton, *Some recent research in the analysis of mixture distributions*, Statistics 21 (1990), pp. 619–641.
- [55] R.J. Vandenberg and C.E. Lance, *A review and synthesis of the measurement invariance literature: Suggestions, practices, and recommendations for organizational research*, Organ. Res. Methods 3 (2000), pp. 4–70.
- 1056 [56] M. Wedel and W.A. Kamakura, *Market Segmentation: Conceptual and Methodological Foundations*, 2nd ed., Kluwer Academic Publishers, Boston, 2000.
- [57] C.E. Werts, R.L. Linn, and K.G. Jöreskog, *Intraclass reliability estimates: Testing structural assumptions*, Educ. Psychol. Meas. 34 (1974), pp. 25–33.
- [58] J. Williams, D. Temme, and L. Hildebrandt, *A Monte Carlo study of structural equation models for finite mixtures*, SFB 373 Discussion Paper 2002-48, 2002.

1061

1066

1071

1076

1081

1086

1091

1096